

SHAHNAM HOSSAIN JILAN

AI Audio Systems Architect | Founder, iHack Audio

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SUMMARY

Audio production professional with 8 years of experience and 10,000+ hours of critical listening. Built an end-to-end AI audio production platform from zero budget — multi-agent orchestration, semantic audio editing, forensic quality audit, 3D spatial mapping. Published output ranked as a top AI podcast on Apple Podcasts. Deep understanding of both audio production and the AI systems that produce audio.

CORE EXPERTISE

- Audio production (8 years)
- Vocal performance analysis
- Semantic audio validation
- Broadcast-standard mastering
- AI-native pipeline design
- Waveform & spectral analysis
- Voice acoustic fingerprinting
- Multi-agent orchestration

PROFESSIONAL EXPERIENCE

Founder & Lead AI Audio Systems Architect

iHack Audio | 2022 – Present

- Built end-to-end AI audio production platform — 8 modules, 5 orchestrated agents
- Designed kinetic notation system for vocal performance engineering [Archetype, Intensity, Stage]
- Created 5-pass semantic audio editing engine with pre-execution QC (industry first)
- Built 3D spatial audio mapping pipeline targeting A24/Dolby Atmos standard
- Published output ranked as top AI podcast — zero marketing spend

Top Rated Audio Editor & Podcast Producer

Upwork | 2017 – Present

- 900+ production hours logged. Top Rated status maintained.
- Podcast production, audiobook narration QC, sound design, professional mastering
- Multi-character dialogue production for international clients

IHACK AUDIO PLATFORM (V3.0.0)

- Multi-Agent Swarm: 5 orchestrated agents (Jarvis, Aura + workers), voice-native control
- Audio Studio Pro: Semantic editing — Whisper → word-level timestamps → VAD → Gemma analysis → 5-pass validation
- QuadCore: 4 parallel chapter engines, up to 30 API keys distributed across pool
- Forensic Audit System: Automated quality scoring — scripting, technical, vocal, marketability
- 3D Spatial Map Engine: Line-by-line spatial audio design with stereo field coordinates
- Maya LoRA Voice Lab: Acoustic fingerprinting — F0, RMS, 13 MFCCs, cosine similarity scoring
- Jojo: Voice-native controller — entire app operable by speech via Gemini Live API

BUILT ON ZERO

Total infrastructure budget: \$0. Every system built on free tier — Google AI Studio, Groq, OpenRouter, Kaggle free GPU. Semantic editing engine runs on Kaggle notebook with live tunnel. Rate limits, quota caps, connection drops — all solved in architecture. Output still scored Tier 1 Broadcast standard.

TECHNICAL STACK

Audio: Adobe Audition, iZotope RX, Auphonic, Descript, Riverside, Audacity

AI: Gemini 2.5 Pro/Flash, Gemini Live API, Gemma, Groq, Whisper, OpenRouter

Custom: iHack Audio Studio Pro, iHack Mastering App, Maya LoRA Voice Lab, Multi-Agent Swarm

Principles: RMS/LUFS standards, VAD segmentation, spatial audio, kinetic notation, spectral analysis

PUBLISHED WORK

- iHack Audio Podcast — Ranked as top AI podcast on Apple Podcasts
- Multi-character AI narration, cinematic sound design, broadcast-standard mastering
- Production time: 4-6 minutes per episode, publication-ready